Shashank Galla

TRUSTWORTHY INDUSTRIAL AI

Advancing AI for manufacturing quality, reliability, and resilience

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Summary

I develop AI methods that improve quality and reliability in manufacturing.
My research bridges AI Methodology and Manufacturing Science, advancing both intelligent algorithms and our understanding of physical processes across four areas:

Anomaly Detection: Can AI catch rare defects and predict surface quality in precision manufacturing?
→ JMP '25 🏆, JMS '24, Mfg. Letters '26, FST '24, NeurIPS '24 (D&B), ICML-W '24, NeurIPS-W '24 & '25

Synthetic Data Generation: Can AI generate realistic, physics-consistent process data when real data is scarce?
→ NeurIPS '26 (Main Track), CVPR-W '26

Uncertainty Quantification: Can we know how far to trust what AI measures?
→ WACV '27 (In Review)

Manufacturing Cybersecurity: Can AI keep manufacturing systems resilient against cyberattacks?
→ CIRP CMS '26

Research Focus: Computer Vision, Generative AI, Trustworthy AI, Reinforcement Learning, Process Monitoring
Application Domains: Nuclear fusion capsule polishing, semiconductor manufacturing, machine tool cybersecurity

Education

Texas A&M University — College Station, TX

Wm Michael Barnes '64 Department of Industrial and Systems Engineering

Ph.D. in Industrial and Systems Engineering — Jan. 2024 – Present

- *Concentration*: Data Science and Advanced Manufacturing — GPA: 4.0/4.0
- *Advisor*: Dr. Satish Bukkapatnam

Master of Science (Thesis) in Industrial and Systems Engineering — Aug. 2021 – Dec. 2023

- *Thesis*: Detecting anomalous motions during polishing process using sensor fusion and machine learning
- *Committee*: Dr. Satish Bukkapatnam (chair), Dr. Rui Tuo, and Dr. Chabum Lee
- *Certification*: Data and Analytics — GPA: 3.8/4.0

National Institute of Technology Calicut — Kerala, India

Department of Mechanical Engineering

Bachelor of Technology in Industrial (Production) Engineering — Jul. 2016 – Jun. 2020

- *Thesis*: Feasibility study for the adoption of electric vehicles and establishing charging points in India

Publications

* denotes equal contribution

Journal

J4. **Pit Density Forecasting for Abrasive Finishing Processes of Nuclear Fusion Target Capsules**
S. Galla, V. Sunkara, S. M. Hayes, M. Biener, S. Bhandarkar, S. Bukkapatnam
Mfg. Letters '26 (Accepted) *Manufacturing Letters* [Link]
→ *Presented at:* SME NAMRC-54 (Oral)

J3. Outstanding Paper Award Fast-Tracked
Efficient Screening of Rare Large Pit Anomalies on Polished Surfaces Using a Minimalist Sampling Scheme
S. Galla*, S. Canacoo*, Y. Zhong*, S. C. Nalband, S. Hayes Michael, M. Biener, S. Bhandarkar, S. T. S. Bukkapatnam
JMP '25 *Journal of Manufacturing Processes* 147, 80–87 [Link]
→ *Award:* Outstanding Paper Award in Manufacturing Processes, SME NAMRC-53 (1 of 3 from 269

submissions)
→ *Recognition:* Selected for Best Student Presentation Competition

J2. Fast-Tracked
Detecting Anomalous Motions in Ultraprecision Shell-Polishing Process Combining Unsupervised Spectral-Band Identification and Explainable-AI
S. Galla, A. Tiwari, S. C. Nalband, S. M. Hayes, S. Bhandarkar, S. Bukkapatnam
JMS '24 *Journal of Manufacturing Systems* 75, 278–287 [Link]
→ *Recognition:* Selected for Best Student Presentation Competition

J1. **Precision Polishing of Ablator Capsules via In-Situ Process Monitoring and Machine-Learning-Based Optimization**
A. Tiwari, J. Prasad, S. Jin, ***S. Galla***, B. Botcha, S. Hayes, M. Biener, K. Bhardwaj, S. Bukkapatnam, Y. Ding, A. Antonios, P. Baldi, S. Bhandarkar
FST '24 *Fusion Science and Technology* 81 (3), 219–231 [Link]

Conference (Peer Reviewed)

C4. NeurIPS Main Track
Physics-Conditioned Video Diffusion with Kinematic Priors for Fusion Capsule Polishing
S. Galla, A. Hanchate, M. Biener, S. Bhandarkar, S. Bukkapatnam
NeurIPS '26 (Accepted) *Advances in Neural Information Processing Systems*

C3. **Controller-Level Taxonomy for Resilient Cyber-Physical Manufacturing Systems**
V. Garudkar, A. Hanchate, ***S. Galla***, S. Bukkapatnam
CIRP CMS '26 *59th CIRP Conference on Manufacturing Systems, Procedia CIRP* 146, 875–880 [Link]

C2. NeurIPS Datasets & Benchmarks
Nuclear Fusion Diamond Polishing Dataset
A. Alexos, J. Liu, ***S. Galla***, S. Hayes, K. Bhardwaj, A. Schwartz, M. Biener, P. Baldi, S. Bukkapatnam, S. Bhandarkar
NeurIPS '24 *Advances in Neural Information Processing Systems* 37, 48692–48709 [Link]
→ *Presented at:* NeurIPS 2024 (Poster)

C1. **Analysis of Enablers and Barriers in Adopting Electric Vehicles in India: DEMATEL-ISM Approach**
G. Shashank, D. Sairam, B. R. Reddy, K. Afreed, R. Sridharan
ICSCAN '20 *International Conference on System, Computation, Automation, and Networking* [Link]

Workshop (Peer Reviewed)

W4. **Physics-Conditioned Video Diffusion for Nuclear Fusion Capsule Polishing Process Synthesis**
S. Galla, A. Hanchate, M. Biener, S. Bhandarkar, S. Bukkapatnam
CVPR-W '26 *VISION Workshop, IEEE/CVF Conference on Computer Vision and Pattern Recognition* [Link]

W3. **Generation-Based Multi-Modal Anomaly Detection for Nuclear Fusion Target Polishing**
S. Abdelkarim, A. Alexos, ***S. Galla***, V. Sunkara, K. Bhardwaj, S. Hayes, M. Biener, S. Bukkapatnam, P. Baldi, S. Bhandarkar
NeurIPS-W '25 *Machine Learning and the Physical Sciences Workshop at NeurIPS* [Link]

W2. **Domain-Adaptive ML for Surface Roughness Predictions in Nuclear Fusion**
S. Galla, A. Alexos, J. Yoo, J. Liu, S. Hayes, K. Bhardwaj, M. Biener, P. Baldi, S. Bukkapatnam, S. Bhandarkar
NeurIPS-W '24 *Machine Learning and the Physical Sciences Workshop at NeurIPS* [Link]

W1. **Scalable Anomaly Detection in Batch Polishing Processes for Inertial Confinement Fusion Shells**
S. Galla, A. Tiwari, K. Bhardwaj, S. M. Hayes, S. T. S. Bukkapatnam, S. Bhandarkar
ICML-W '24 *AI for Science Workshop at ICML* [Link]

In Review

R1. **Uncertainty Propagation from Detection through Tracking to Orbital Speed in Inertial Confinement Fusion Capsule Polishing**
S. Galla, H. Tian, V. Sunkara, K. Bhardwaj, S. Hayes, M. Biener, S. Bhandarkar, S. Bukkapatnam
Under Review at WACV '27 *IEEE/CVF Winter Conference on Applications of Computer Vision*

Honors and Awards

- **Finalist**, INFORMS QSR Best Refereed Paper Competition — Sep. 2026
- **Honorable Mention**, Titbit Session, 2026 Zorich Workshop on Applied Statistics & AI — Sep. 2026
- **Student Ambassador Scholarship (Generative AI Track)**, TAMU Institute of Data Science — FY. 2026-27
- **Outstanding Paper Award in Manufacturing Processes**, SME NAMRC-53 — Jun. 2025
  (One of three papers selected from 269 paper submissions)
- **Runner-up**, IISE QCRE Data Challenge — May 2025
- **Finalist**, Best Student Research Presentation, SME NAMRC (52 & 53) — Jun. 2024 & 2025
- **Finalist**, IISE DAIS Data Challenge — May 2024
- **Travel Grant Award**, Department of Industrial and Systems Engineering — 2024, 2025, 2026
- **First Place**, Baker Hughes Data Challenge at TAMU Datathon — Oct. 2022

Teaching

- **TA** of ISEN 311: Introduction to Data Analytics and Engineering (17 undergraduate students) — Spring 2025
- **TA** of DAEN 323: Statistical Learning and Decisions (36 undergraduate students) — Spring 2025

Service

- **Reviewing**
  - **Journal:** IEEE Transactions on Semiconductor Manufacturing, International Journal of Mechanical Sciences
  - **Conference and Workshops:** NeurIPS (2026, 2024), WACV (2024), NeurIPS ML for Physical Sciences Workshop (2025, 2024), NeurIPS Foundation Models for Science Workshop (2025, 2024), ASME IMECE (2025), IEEE CASE (2025), AAAI (2026, 2025)
- **Leadership**
  - **President**, INFORMS Student Chapter at TAMU — Sep. 2025 – Present
  - **Organizer**, NewInML Affinity Workshop at NeurIPS — Dec. 2025 & 2026
  - **Vice President** of Academic Affairs, INFORMS Student Chapter at TAMU — Apr. 2025 – Aug. 2025
  - **Director of Events**, TATHVA and RAGAM, UG technical and cultural fests — Oct. 2018 – Jan. 2020

Talks and Posters

- Forecasting Pit-Distribution for Rare-Defect Screening in Nuclear Fusion Target Capsule Finishing Operations
  - *INFORMS Annual Meeting, San Francisco, CA, QSR Best Refereed Paper Finalist* — *Upcoming*
- Uncertainty Propagation from Detection through Tracking to Orbital Speed in Inertial Confinement Fusion Capsule Polishing
  - *2026 Zorich Workshop on Applied Statistics & AI, Titbit Session (Flash presentation)* — *Sept. 2026*
  - *Guest Lecture for the Computer Vision for Manufacturing class at Virginia Tech* — *Apr. 2026*
- Pit Density Forecasting for Abrasive Finishing Processes of Nuclear Fusion Target Capsules
  - *SME NAMRC-54, State College, PA, Oral Presentation (Manufacturing Processes XI session)* — *Jun. 2026*
- Physics-Conditioned Video Diffusion for Nuclear Fusion Capsule Polishing Process Synthesis
  - *CVPR 2026 VISION Workshop, Denver, CO, Poster* — *Jun. 2026*
- Motion analysis of nuclear fusion target capsules during ultra-precision polishing
  - *IISE Annual Conference, Oral Presentation* — *May 2026*
- Efficient screening of rare large pit anomalies on polished surfaces using a minimalist sampling scheme
  - *Invited talk by the Society of Manufacturing Engineers for the Manufacturing Edge Webinar* — *Apr. 2026*
  - *NAMRC-53 Conference, Presentation Competition* — *Jun. 2025*
- How AI Meets the Physical World: Engineering the Future from Sensors to Intelligence
  - *Invited talk at Sreenivasa Institute of Technology and Management Studies, AP, India* — *Jan. 2026*

- End-to-end resilience for machine tool controllers against cyberattacks
  - *INFORMS Annual Meeting, Atlanta, GA, Invited Presentation* *Oct. 2025*
- Nuclear Fusion Diamond Polishing Dataset
  - *Neural Information Processing Systems Conference: Datasets and Benchmarks Track, Poster* *Dec. 2024*
- Domain-Adaptive ML for Surface Roughness Predictions in Nuclear Fusion
  - *Neural Information Processing Systems Conference: ML4PS Workshop, Poster* *Dec. 2024*
- Detecting anomalous motions in ultraprecision shell-polishing combining unsupervised spectral-band identification and X-AI
  - *NAMRC-52 Conference, Presentation Competition* *Jun. 2024*
  - *Los Alamos National Laboratory Executive Visit at TAMU, Poster* *Feb. 2024*
  - *TAMU ERS Energy Conference, Oral Presentation* *Oct. 2023*
  - *IISE Annual Conference, Oral Presentation* *May 2023*

Industry Experience

Seagate Technology Bloomington, MN

Machine Learning Engineering Intern: Wafer Process Development May 2026 – Aug. 2026

- Developed a scalable framework for generating controllable synthetic defects, enabling robust anomaly detection across multiple semiconductor manufacturing stages with limited real-world defect data
- Leveraged foundation models to build a unified anomaly detection and classification framework, reducing the need for separate task-specific models and improving efficiency for production deployment

Machine Learning Engineering Intern: Seagate Research Group May 2025 – Aug. 2025

- Developed a custom U-Net with adaptive loss functions to recover fine structural details in microscopy images, enabling more precise materials characterization
- Built and benchmarked VAEs, GANs, and diffusion models to generate realistic frequency-response data, supporting data-driven controller design and performance evaluation

Kinaxis Corp. Dallas, TX

Solutions Consultant Intern: Professional Services Team May 2023 – Aug. 2023

- Responsible for building end-to-end demand and supply planning models and providing solution for a bicycle company

Wayfair, LLC Boston, MA

Data Scientist Co-Op: Analytics and Insights Team Jul. 2022 – Jan. 2023

- Responsible for building predictive models, dashboards, creating weekly and monthly business reports for customer service performance improvement teams

Mentoring Experience

- **Graduate** (Texas A&M)
  - **Jaemin Song** (MS IE, Fall 2026) – *Deep reinforcement learning for cyberattack control*
  - **Vikash Sunkara** (MS EE, Summer & Fall 2025) – *Pit defect evolution modeling using flow-based learning*
  - **Vaishnavi Garudkar** (MS IE, Fall 2024) – *Deep RL for cyberattack control* (co-mentored)
  - **Dylan Lasrado** (MS IE, Fall 2024) – *Understanding training dynamics of GAN models*
- **Undergraduate** (Texas A&M unless noted)
  - **Aninditha Sriramasubramanian** (Fall 2026) – *Tracking the motion of capsules during polishing*
  - **Jayin Palicha** (Fall 2026) – *Kinematics-driven synthetic video generation for capsule polishing*
  - **Asvath Madhan** (Fall 2026) – *Velocity estimation of nuclear fusion target capsules*
  - **Pranav Venkatramanan** (Summer & Fall 2025) – *Velocity estimation of nuclear fusion target capsules*
  - **Vraj Patel** (Fall 2023 & Spring 2024) – *Anomaly detection using sensor fusion and machine learning*
  - **Akhilesh Narayan** (IIT Bombay, Fall 2023) – *Detecting shell sliding and slipping with deep learning*

Professional Memberships

IISE | INFORMS | IEEE